

Phase-mismatched waveguide to racetrack-type microresonator coupling

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Abstract

We propose a phase-mismatched coupling scheme between a bus waveguide and a racetrack-type microresonator. The width of the bus waveguide is designed to be much narrower than that of the microresonator, introducing a significant phase mismatch. The coupling region is then extended to allow longer interaction with multiple optical power exchanges. This results in a sinusoidal couplings across the spectrum, enabling periodic critical couplings that are independent of fabrication-induced gap variations.

Device Structure

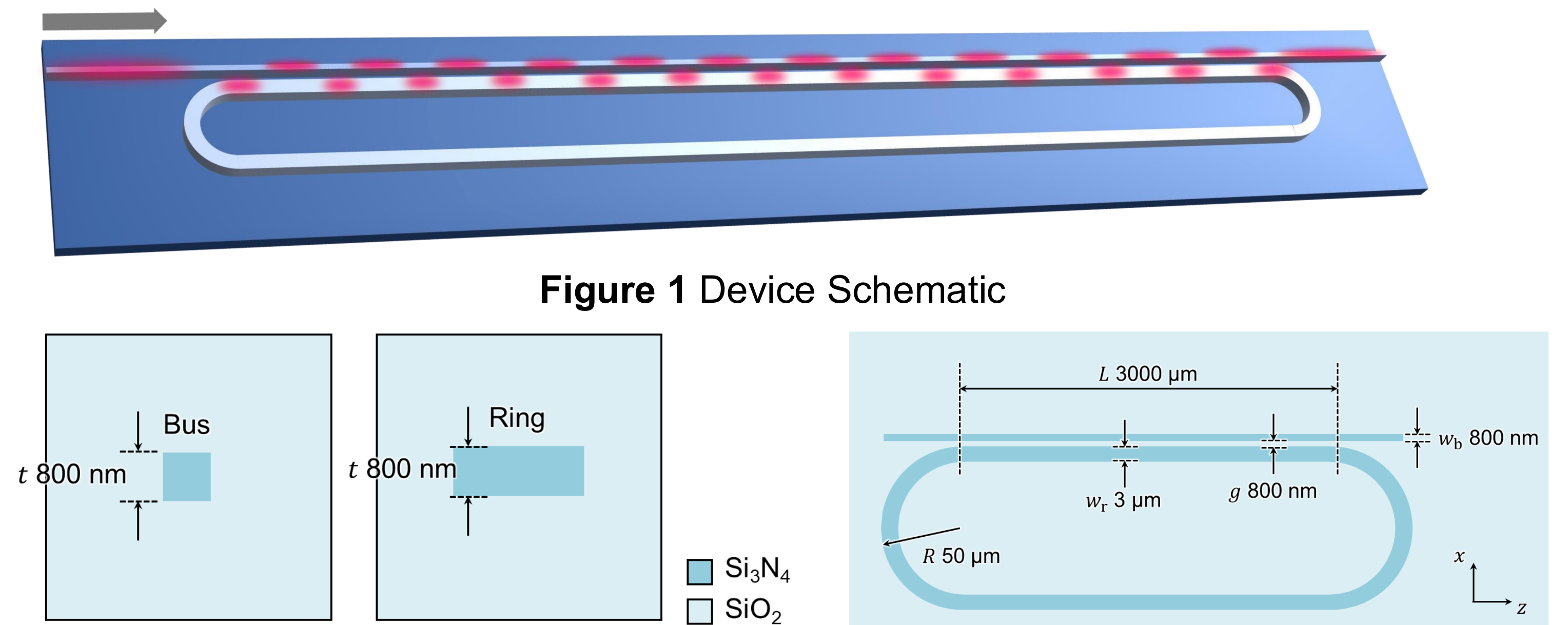


Figure 2 Cross sectional structure

Figure 3 Top-view structure

Device Concept

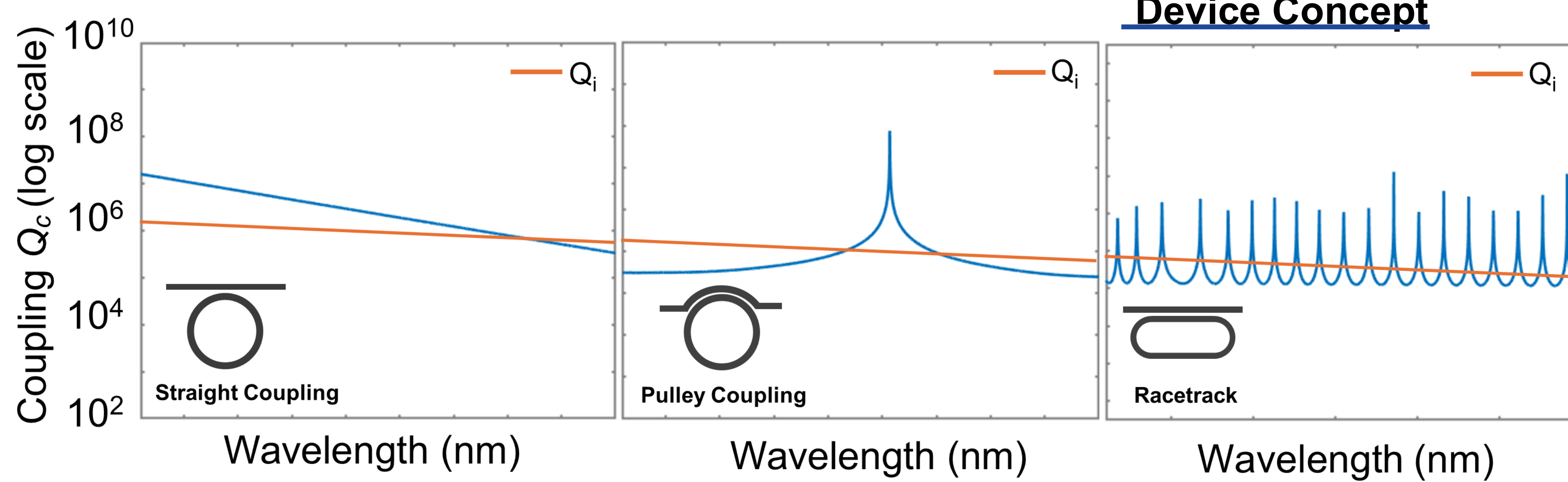


Figure 4 Coupling Characteristics in Three Different Types of Coupling

$$\kappa_{r \rightarrow wg} = \Gamma L_c \text{sinc} \left(L_c \sqrt{(\Delta\beta/2)^2 + \Gamma^2} \right)$$

$$Q_c = \omega \frac{n_g^R}{c_0} \frac{2\pi R}{|\kappa_{r \rightarrow wg}|^2} = \omega \frac{n_g^R}{c_0} L_{\text{roundtrip}} \left[\Gamma L_c \text{sinc} \left(L_c \sqrt{(\Delta\beta/2)^2 + \Gamma^2} \right) \right]^{-2}$$

- Γ : Ring/Waveguide mode overlap [m^{-1}]
- κ : Coupling coefficient [ratio]
- L_c : Coupling region length [m]

Experimental Result

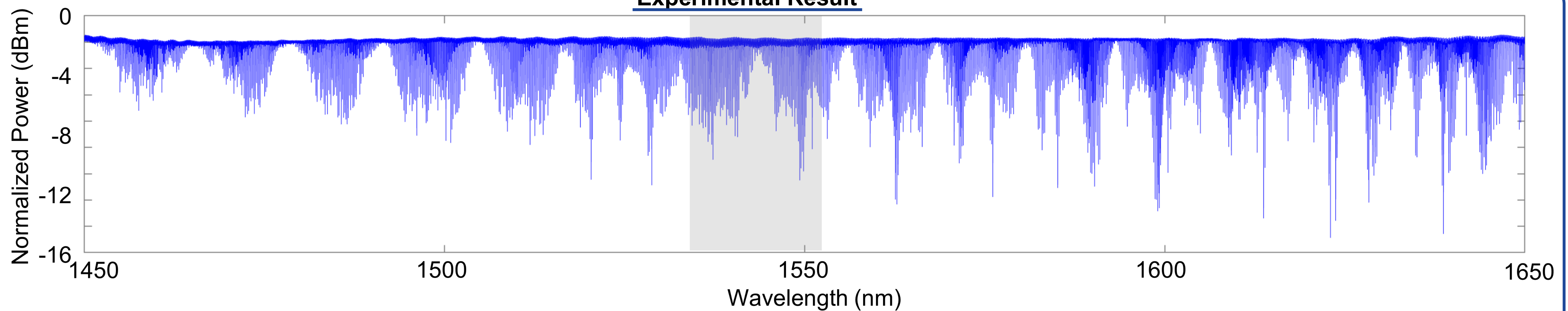


Figure 5 Experimental Transmission data

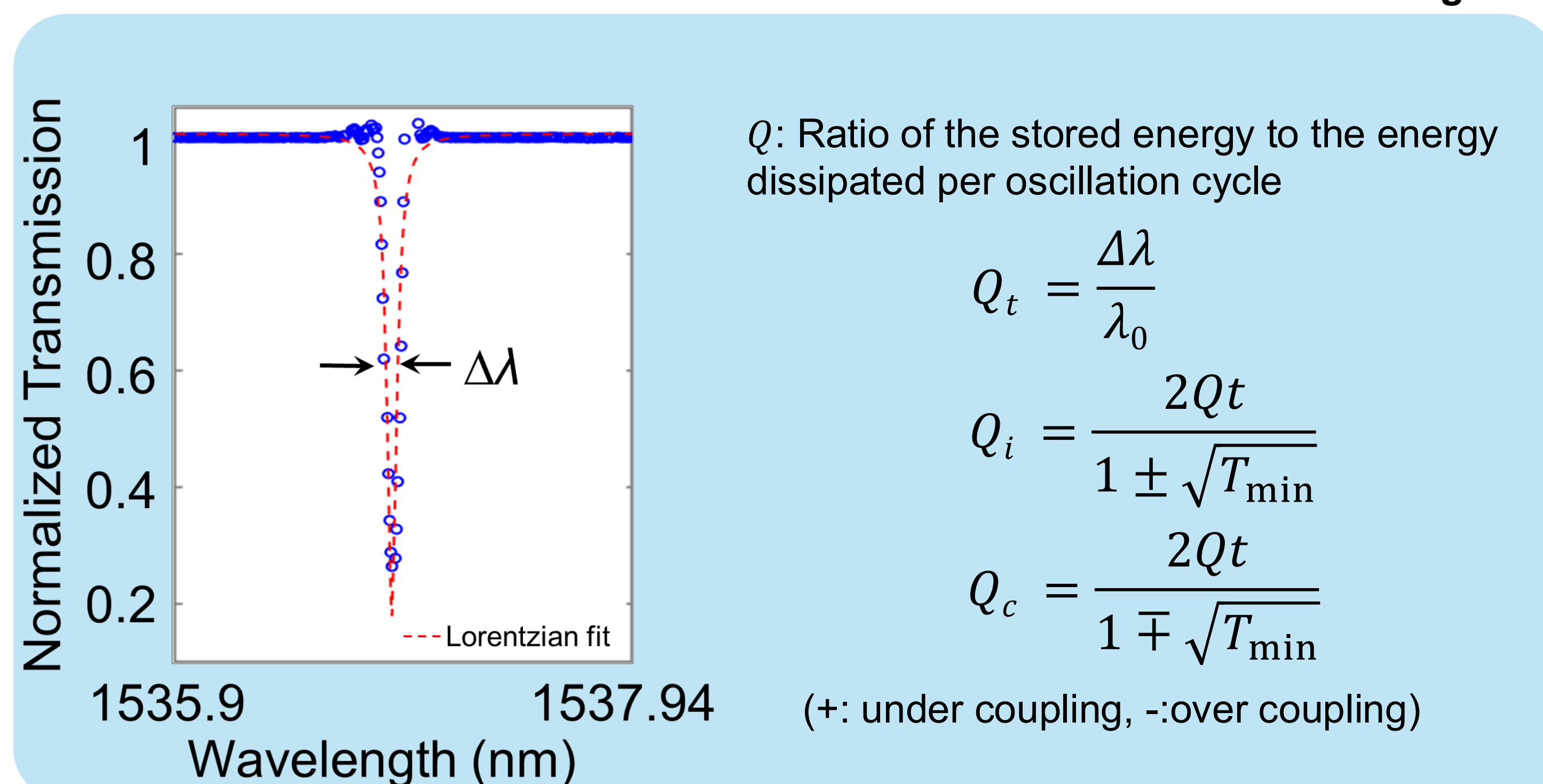


Figure 6 Q fitting spectrum

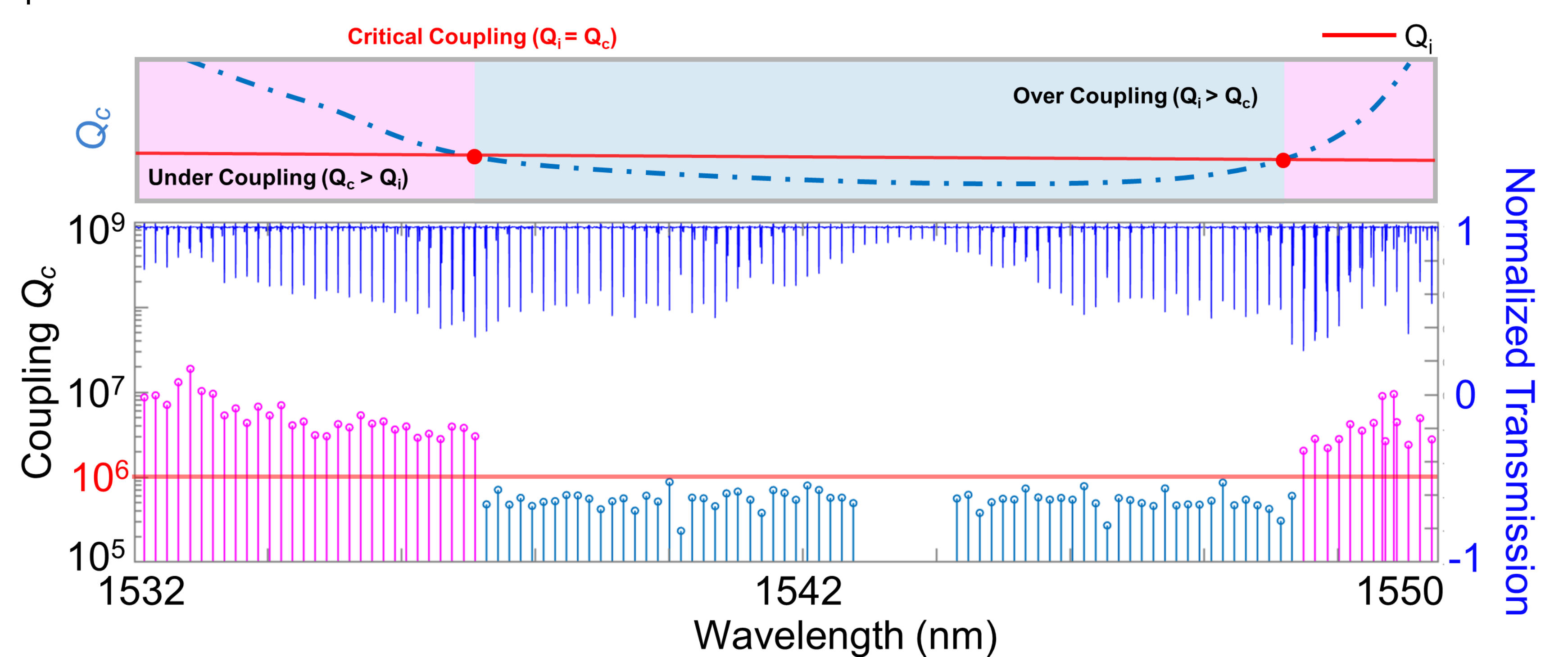


Figure 7 Coupling Q spectrum with transmission

Discussion

- In our approach, we introduced a phase-mismatched waveguide-microresonator coupling concept, in contrast to conventional phase-matched couplings (e.g., straight and pulley couplings).
- Unlike conventional coupling methods, where the phase matching between the bus waveguide and ring is considered ideal, we show that phase mismatching can be rather effective under varied gap profiles.
- Further modeling and optimization can be explored for achieving coupling ideality under phase-mismatched condition.

Conclusion

In this study, we proposed a waveguide-microresonator coupling scheme that utilizes phase-mismatching between them. The experimentally characterized transmission spectrum exhibits a sinusoidal coupling coefficient (κ) across the spectrum, enabling periodic critical coupling points independent of fabrication-induced gap variations.

References

- [1] Moille et al. "Broadband resonator-waveguide coupling for efficient extraction of octave-spanning microcombs." Optics letters 44.19 (2019)
- [2] A. Yariv and P. Yeh, Photonics: Optical Electronics in Modern Communications (Oxford University, 2007).